Online Learning-Based Beamforming for Rate-Splitting Multiple Access: A Constrained Bandit Approach

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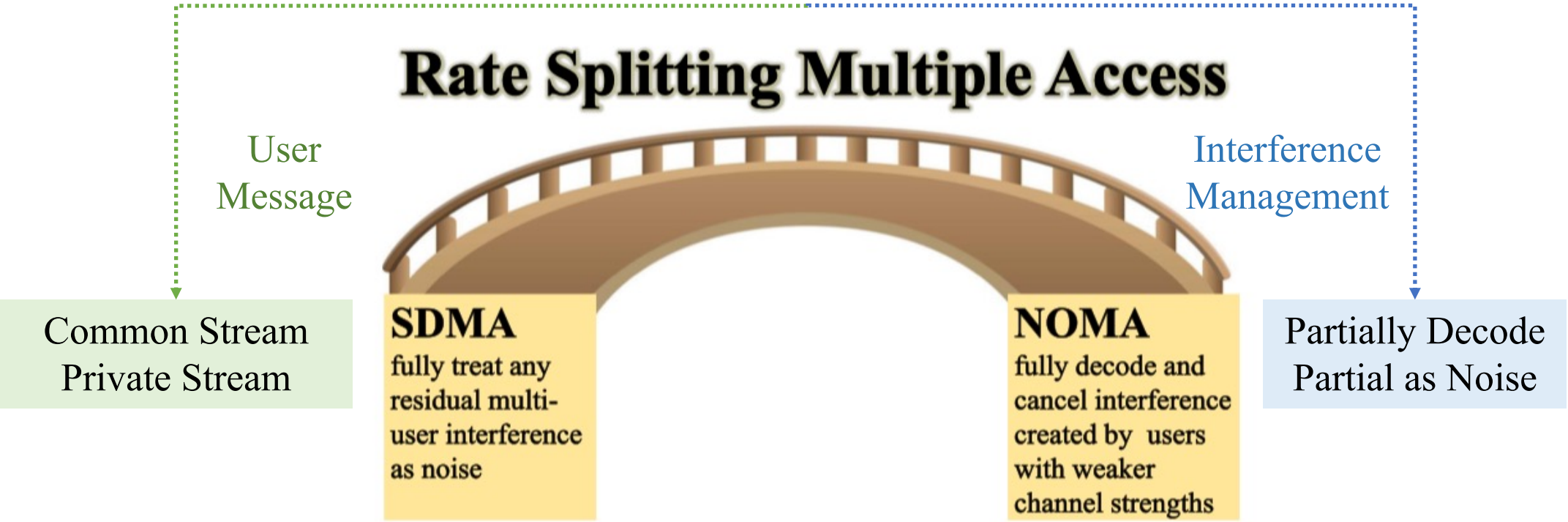
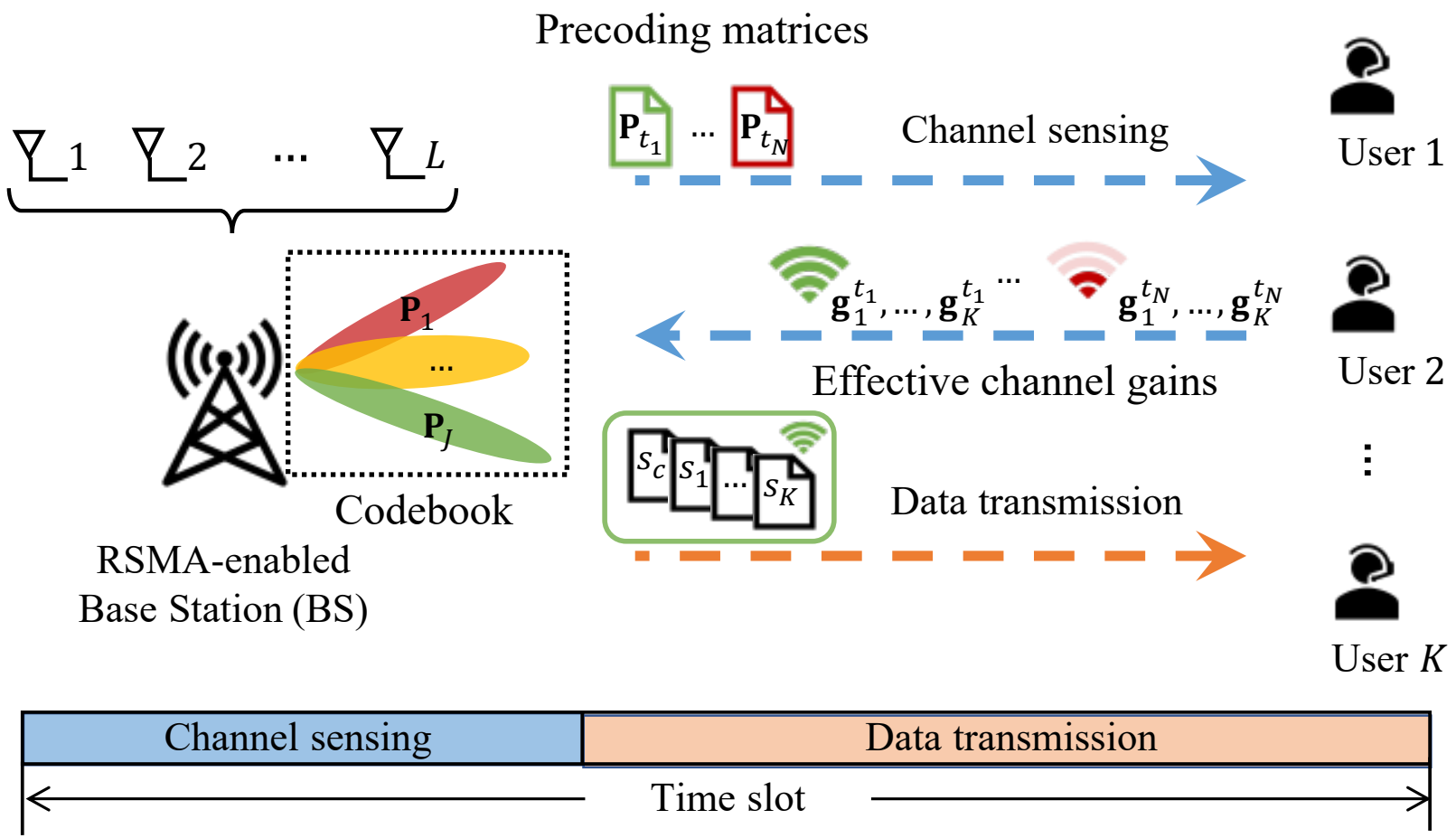
FineComLab

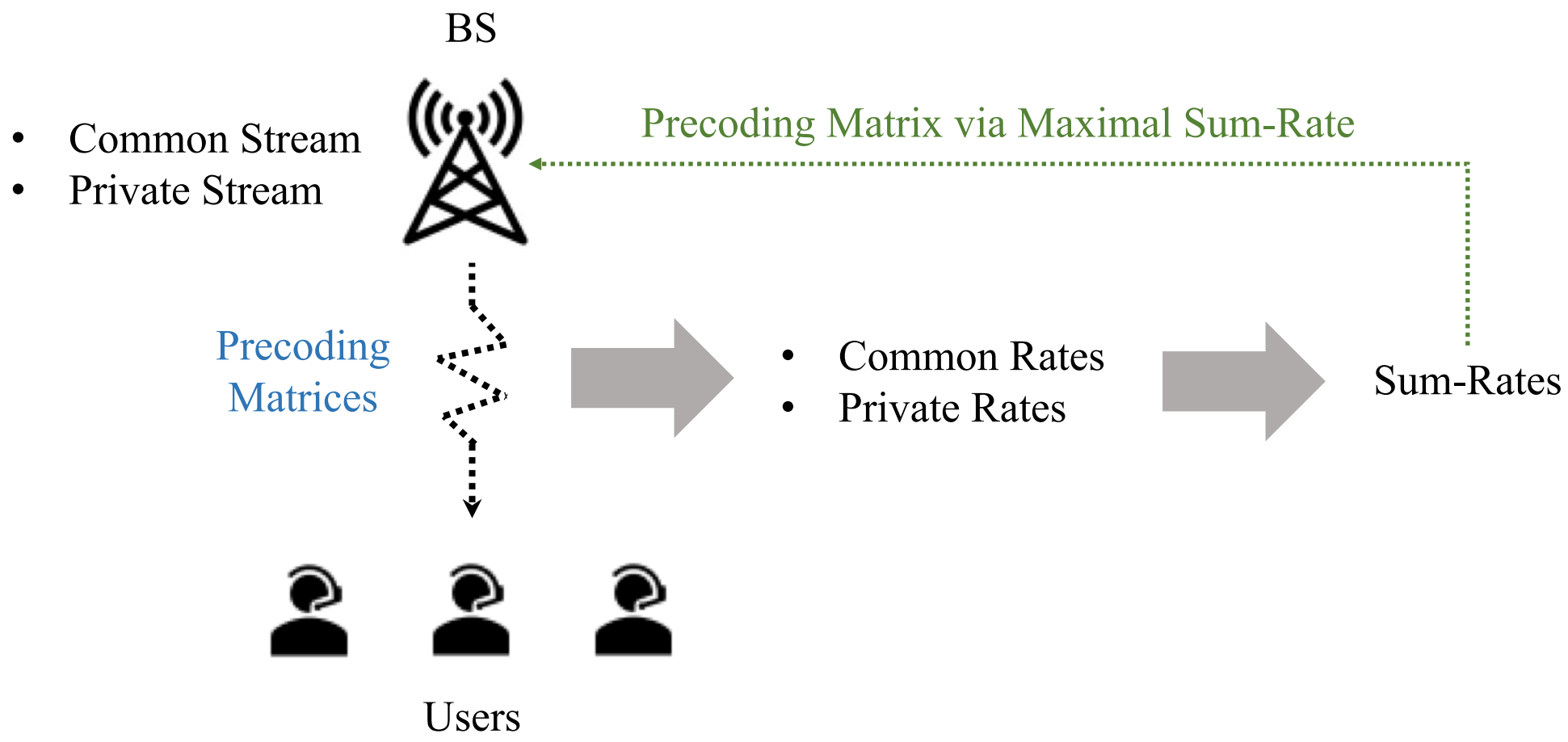
Rate-Splitting Multiple Access (RSMA)



RSMA for Downlink MIMO Systems



Precoder Selection in Channel Sensing



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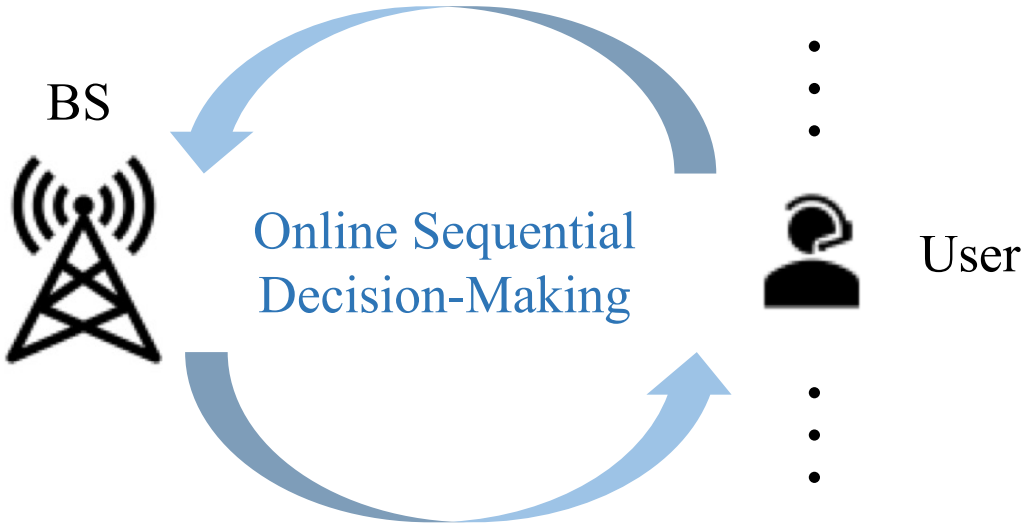
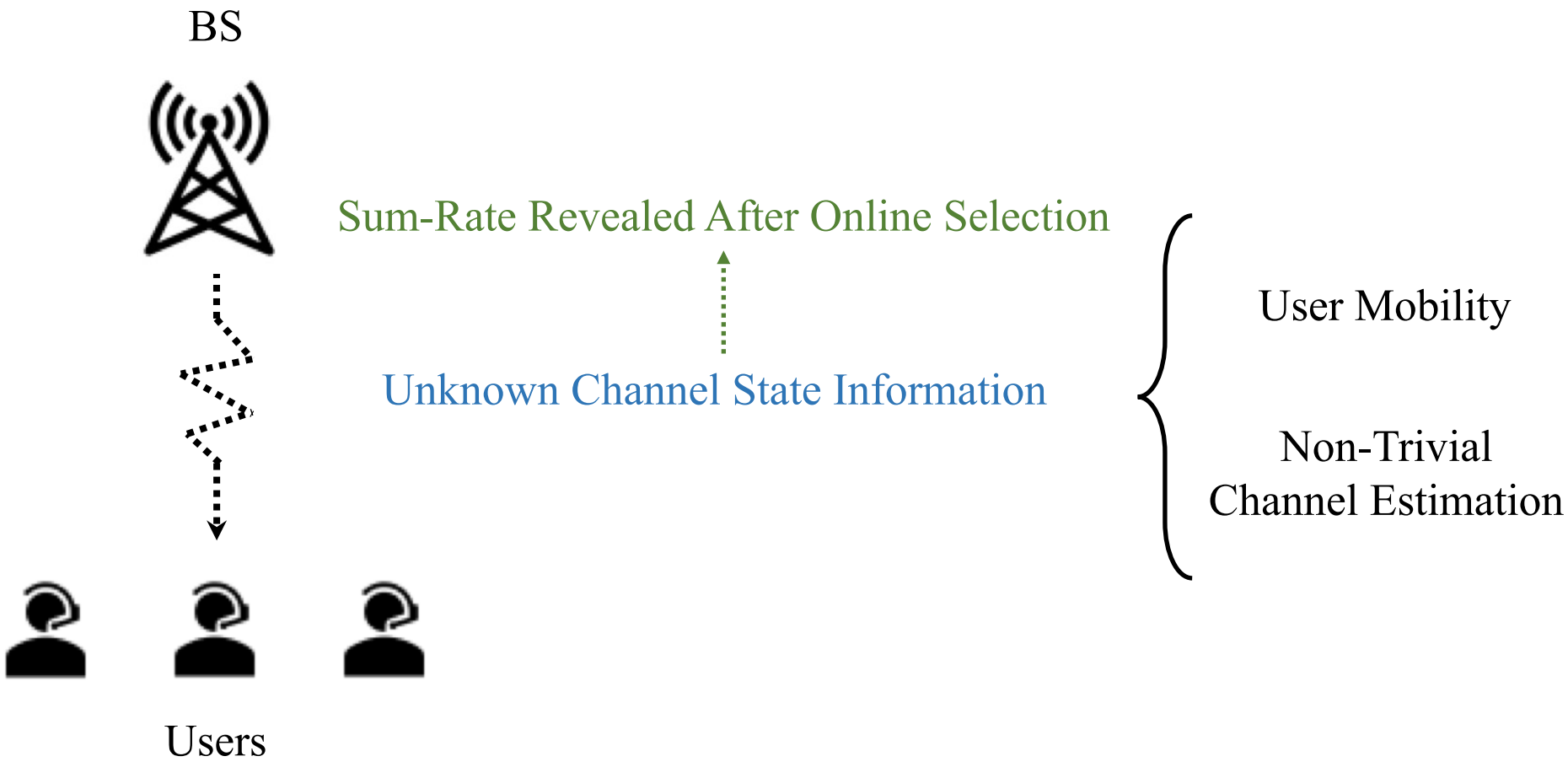


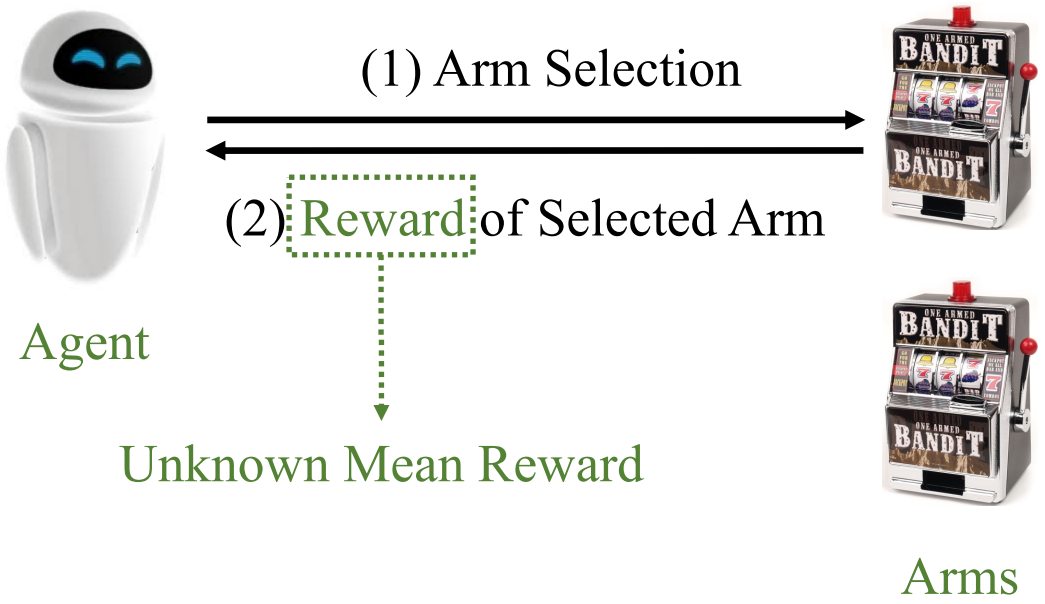
- Sum-Rate Maximization
- Multi-Round Selection

How to Select Precoding Matrices in Each Round?

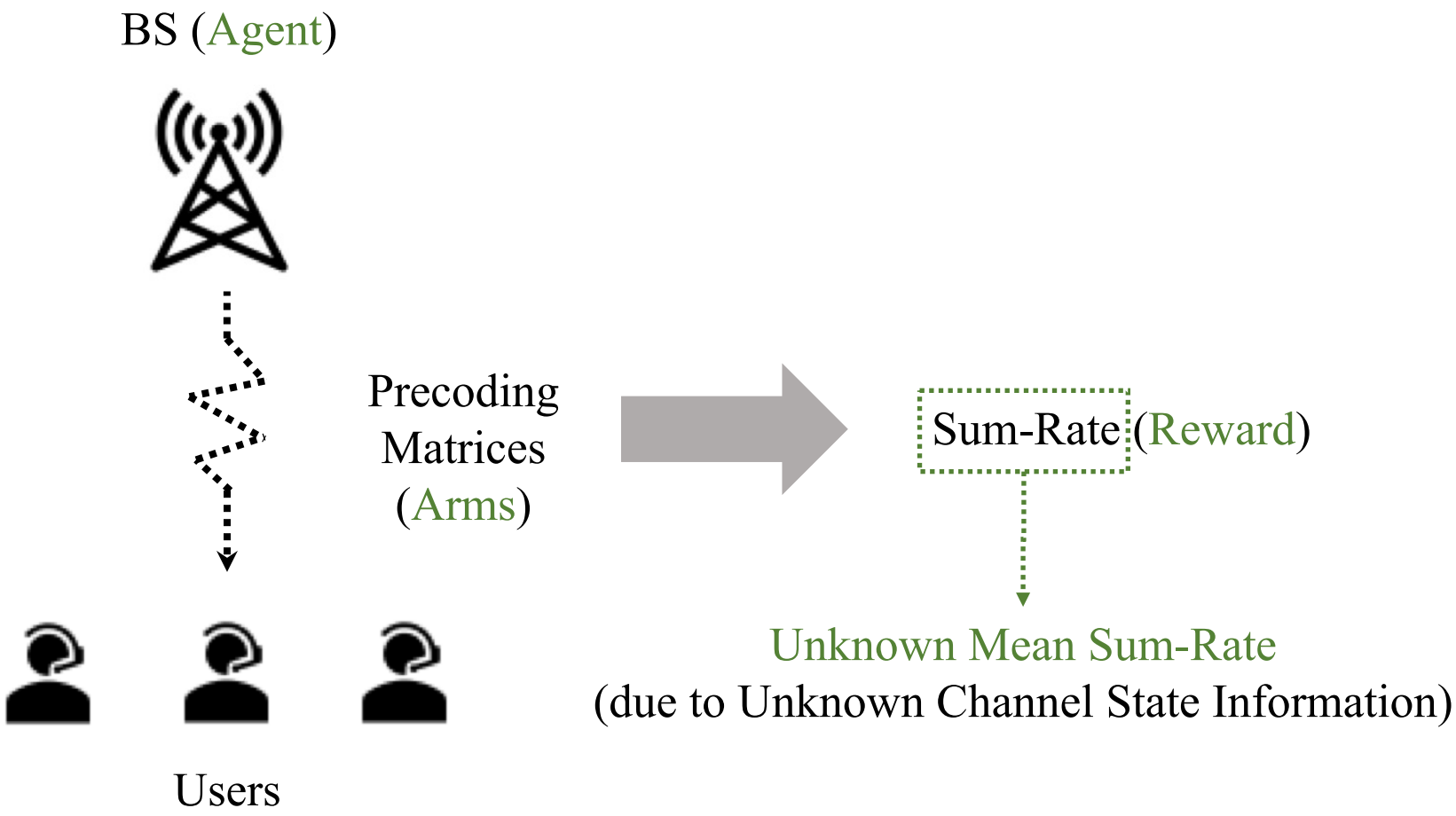
Unavailable Channel State Information at Transmitter



Online Selection via Multi-Armed Bandits

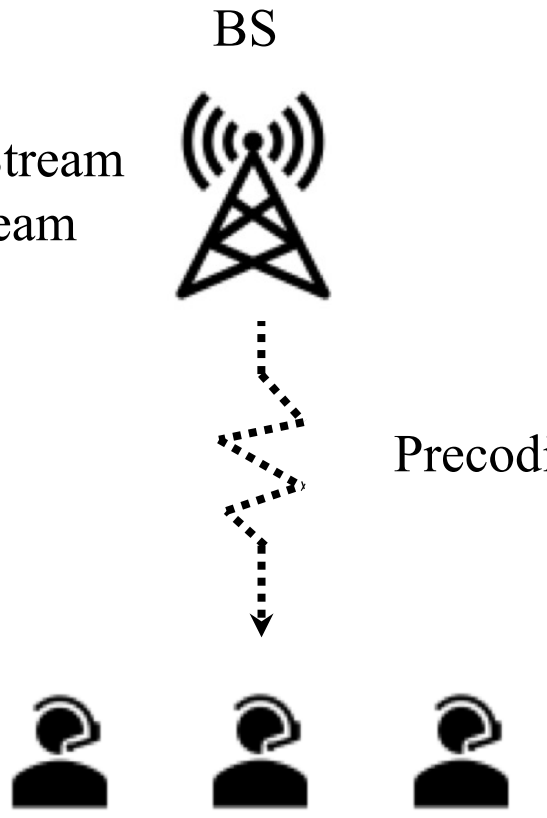


A Bandit Formulation of Precoder Selection



Power Constraint

- Common Stream
- Private Stream



Precoding Matrix

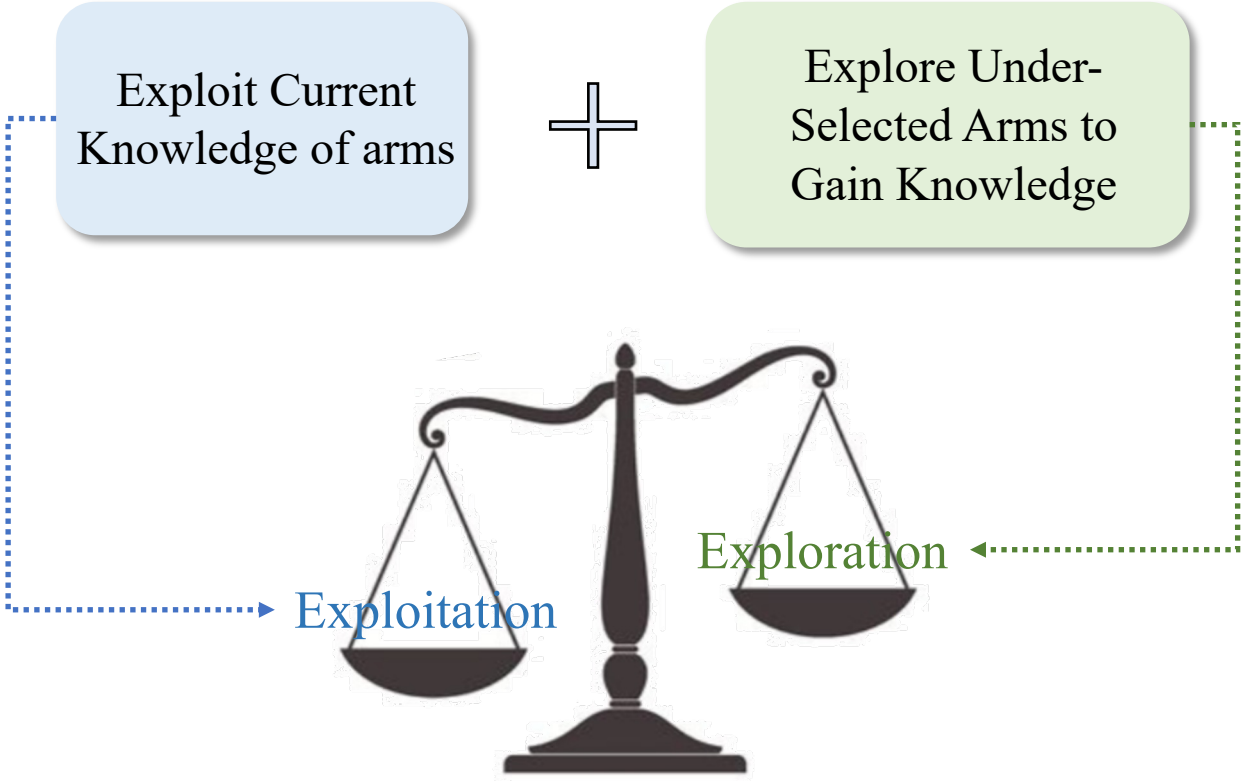
Power Consumption

Low Power Efficiency

Within-Budget Guarantee

Sustainable Deployment

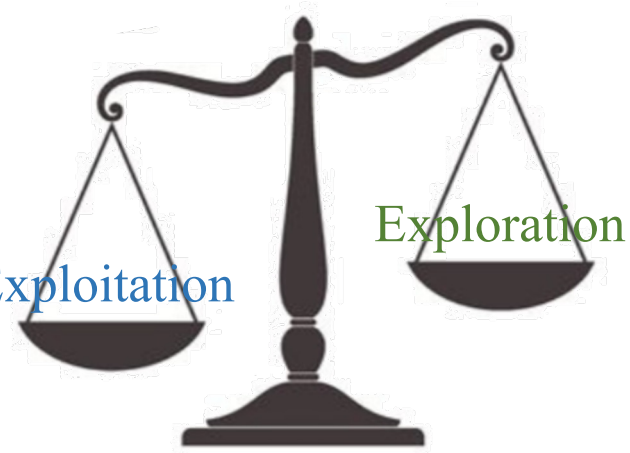
Online Learning for the Unknown Mean Sum-Rate



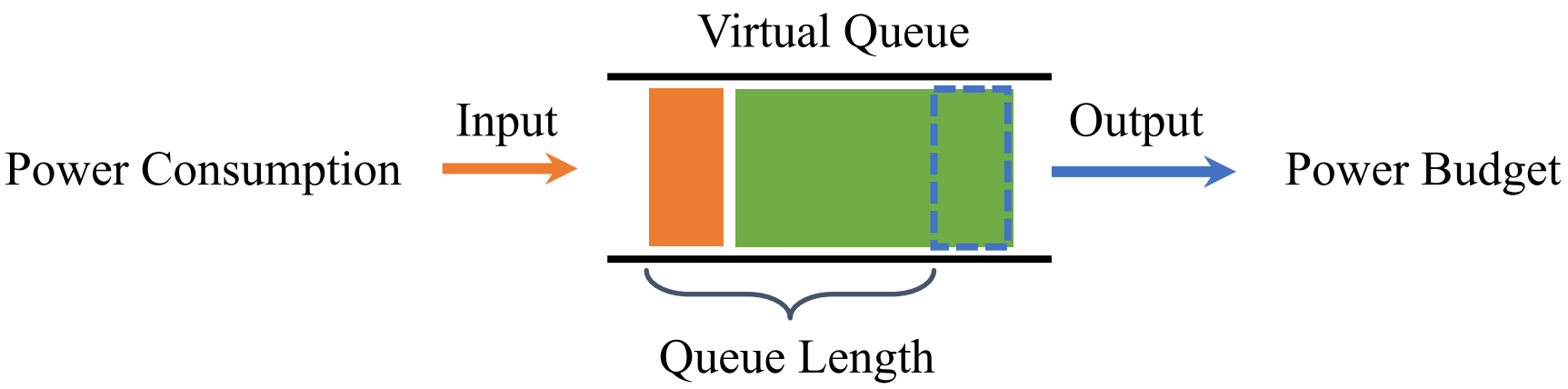
Upper-Confidence-Bound-Style Approach

Confidence Radius

Empirical Mean $+$ $\sqrt{\dfrac{\log t}{\text{Number of Selected Times}}}$



Online Control with Virtual Queue



Constraint Satisfaction ⟷ Controlled Virtual Queue Length

Integrated Online Algorithm Design

Online Learning

Sum-Rate Maximization

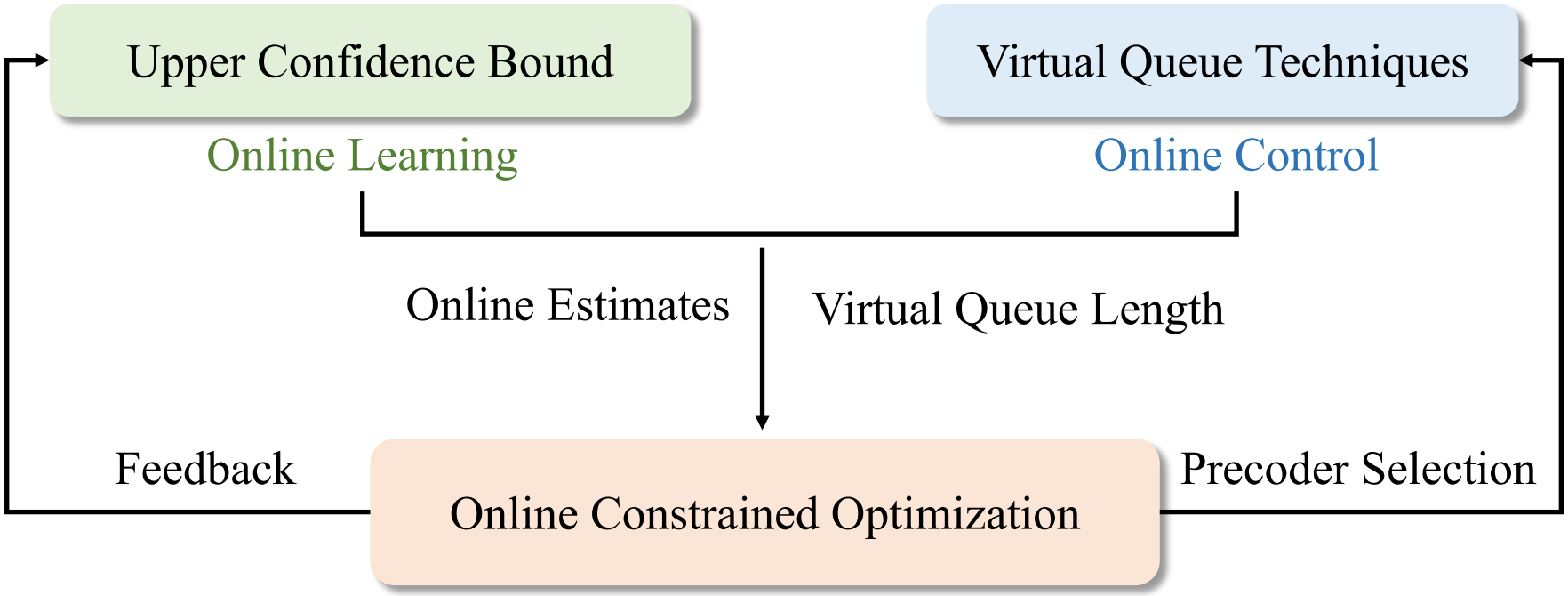
Online Control

Constraint Satisfaction

Maximize $V \times$ Mean Sum-Rate Estimate $=$ Virtual Queue Length $\times$ Power Consumption

Balance Sum-Rate Maximization & Constraint Satisfaction

Algorithm Workflow



Queue Stability

Queue Stability Definition: $\limsup_{T' \to \infty} \frac{1}{T'} \sum_{t=0}^{T'-1} \mathbb{E}[Q(t)] < \infty$

Queue Length of Virtual Queue Q for the Power Constraint

Queue Stability Guarantee: $\limsup_{T' \to \infty} \frac{1}{T'} \sum_{t=0}^{T'-1} \mathbb{E}[Q(t)] \leq \frac{D + V r_{\max}}{\epsilon} < \infty$

Some Finite Constants

Holds with Finite V

Round-Averaged Regret

Sum-Rate Gap from the Optimal

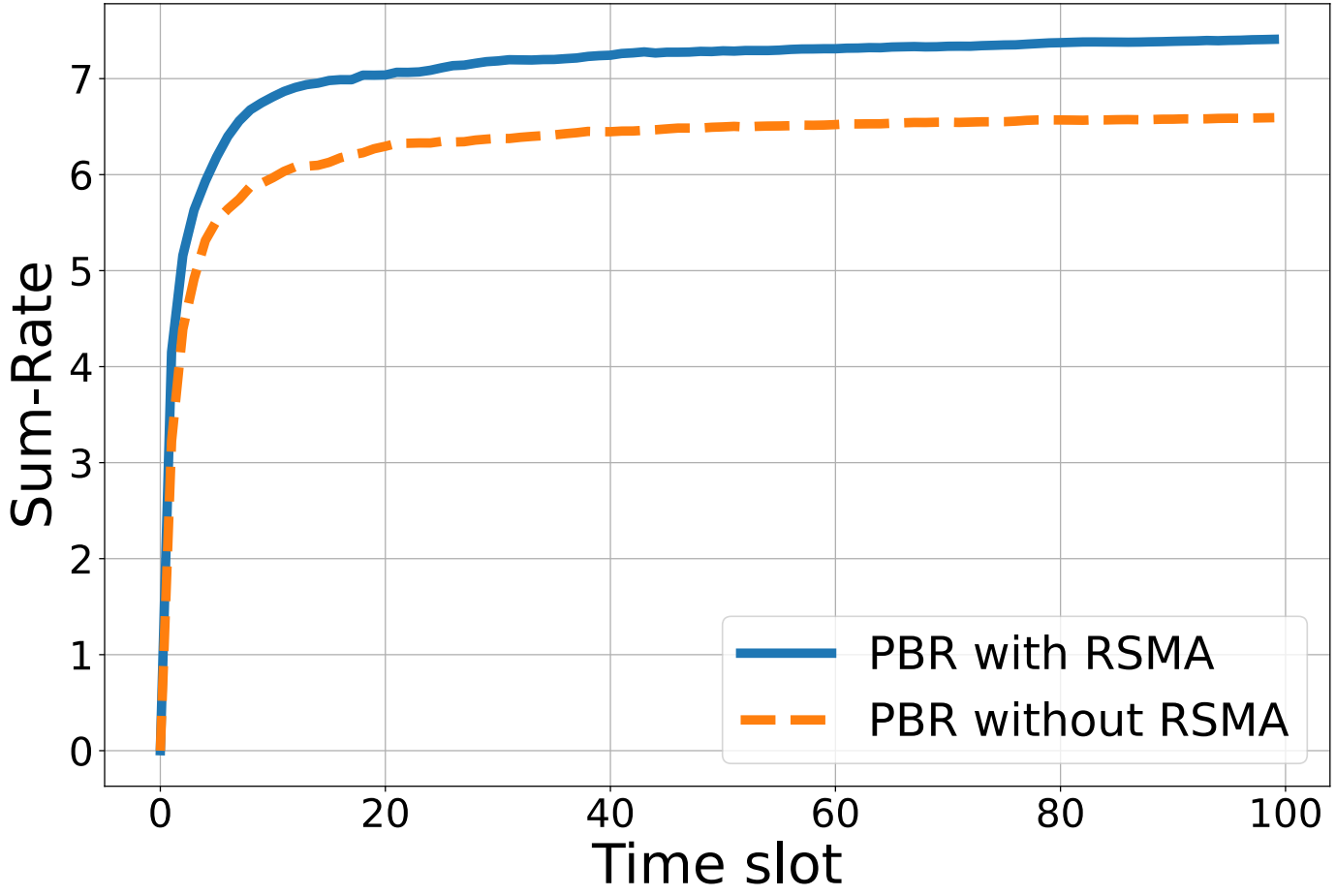
Online Control

Online Learning

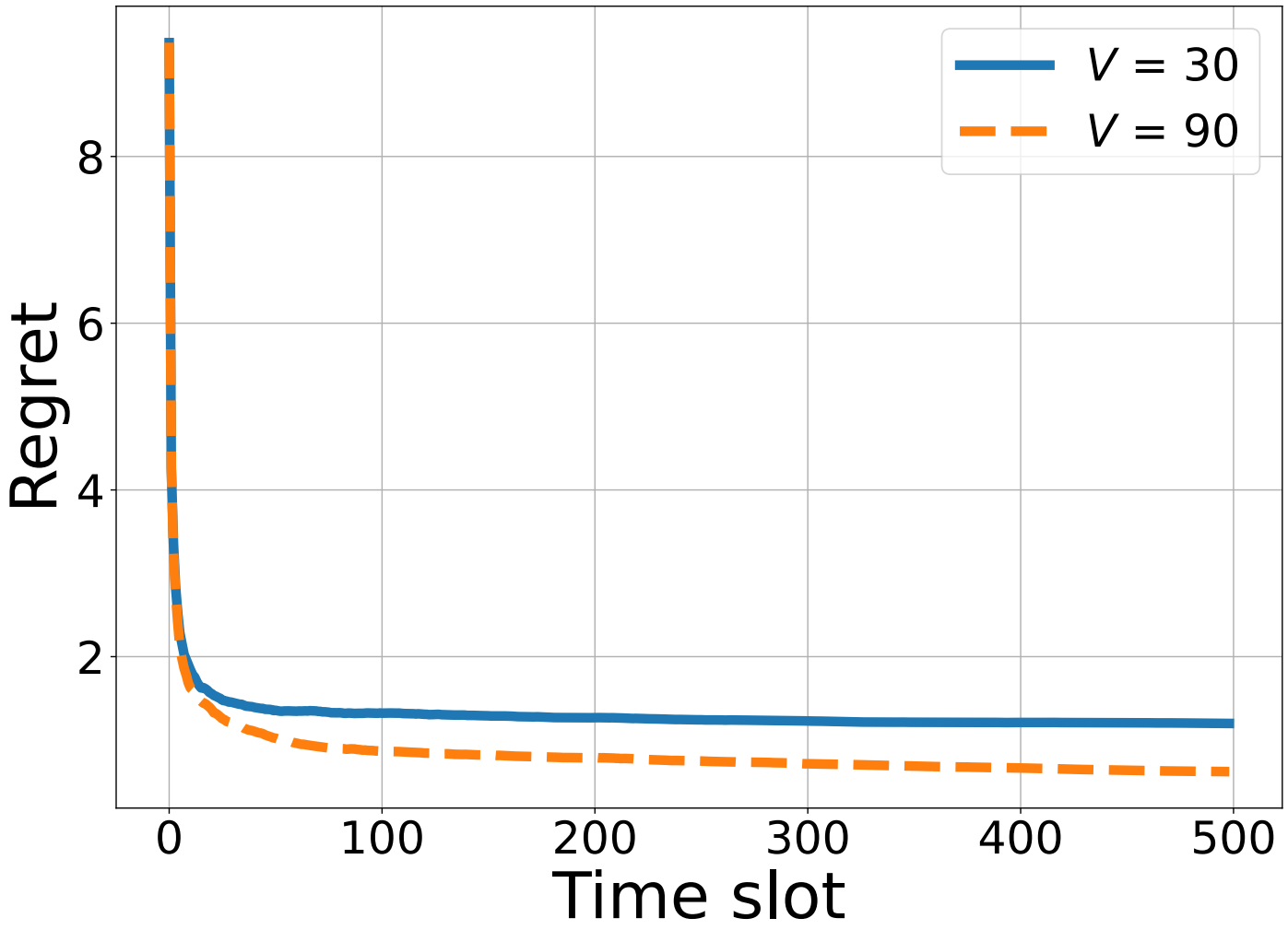
Round-Averaged Regret: $\mathrm{Reg}(T) = O(1/V + \sqrt{\log T / T})$

Sublinear Regret Guarantee: $V = \Omega(\sqrt{T/\log T}) \Rightarrow \mathrm{Reg}(T) = O(\sqrt{\log T / T})$

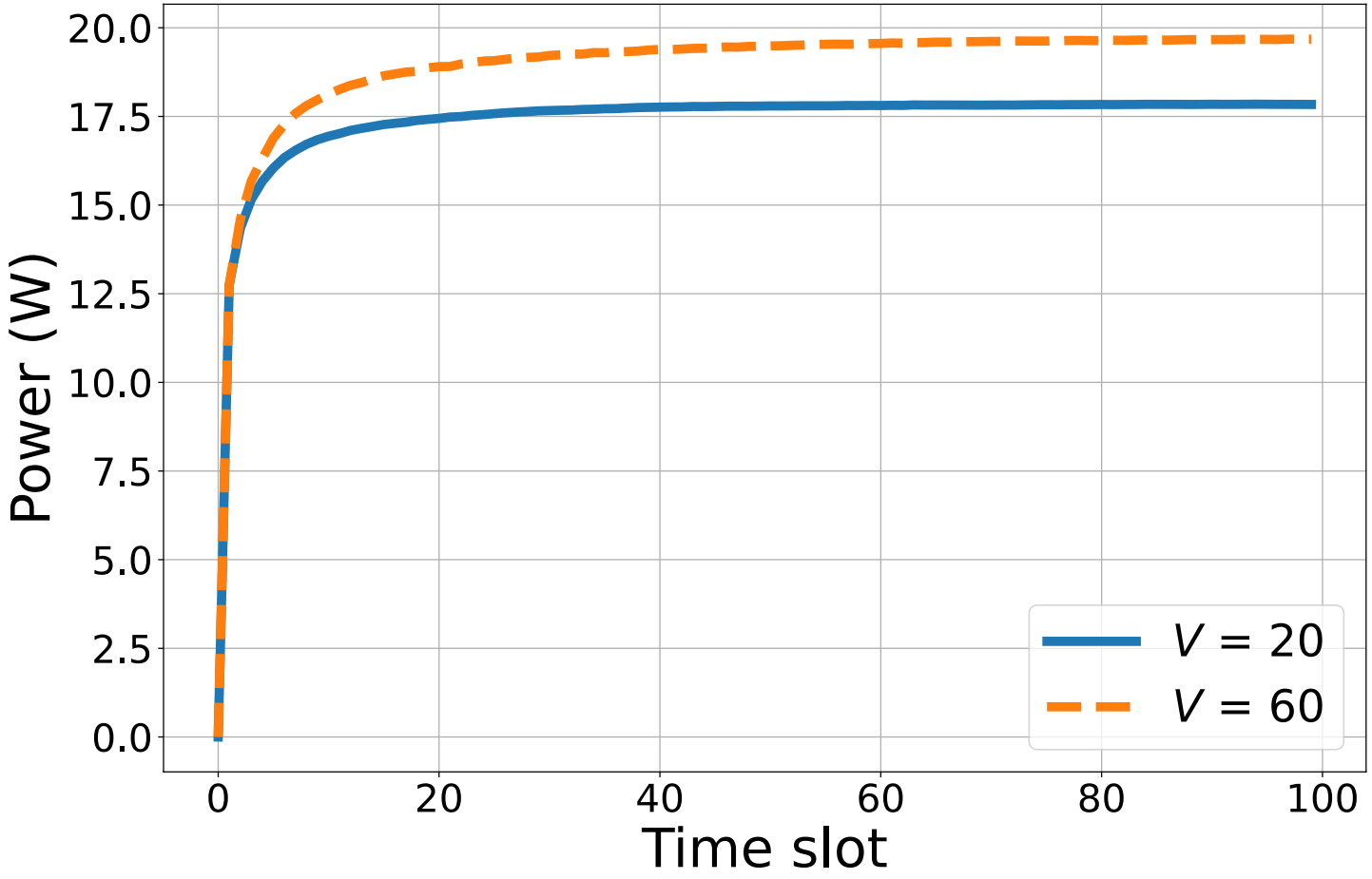
Simulation Result: Sum-Rate with RSMA



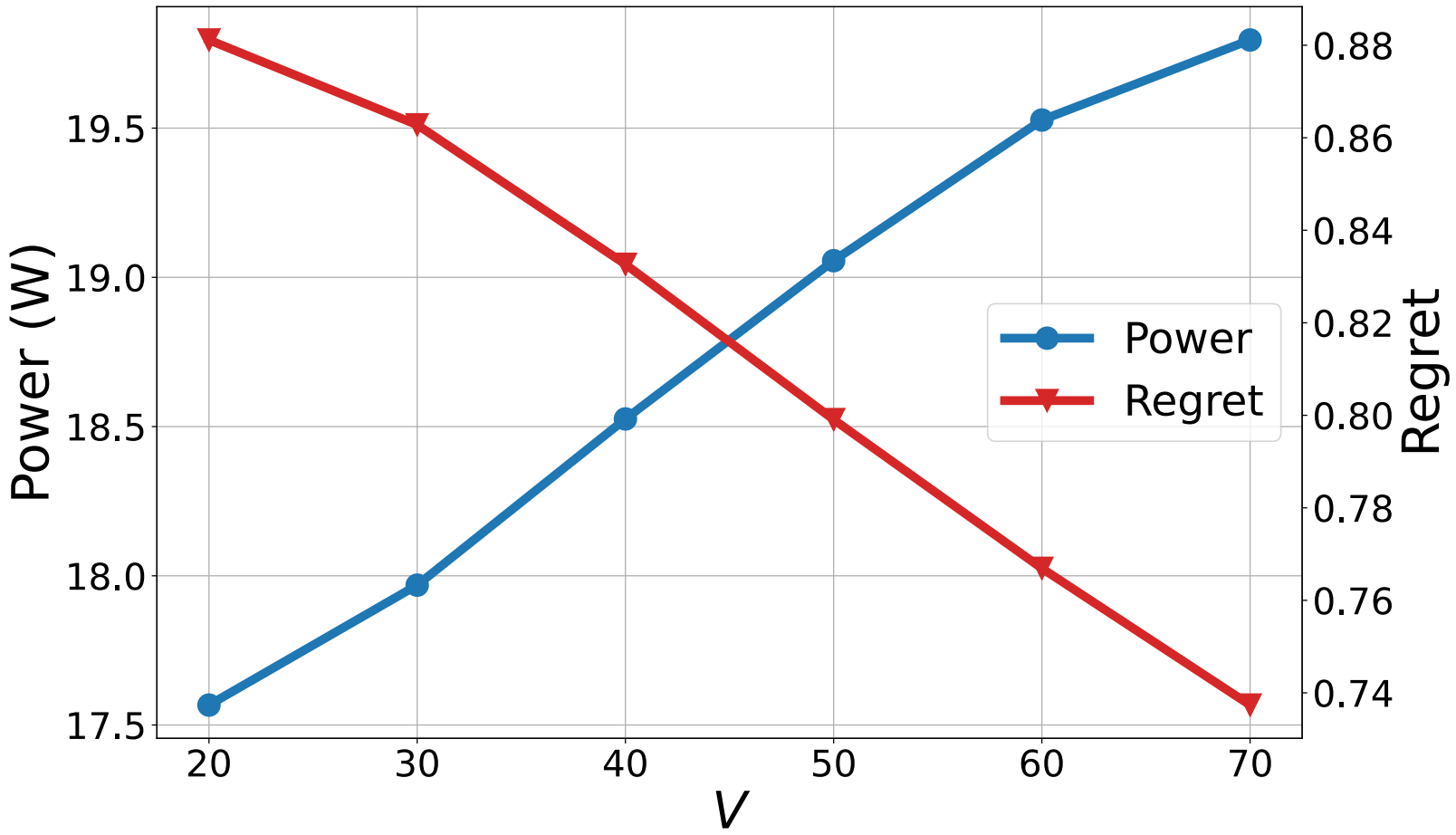
Simulation Result: Regret



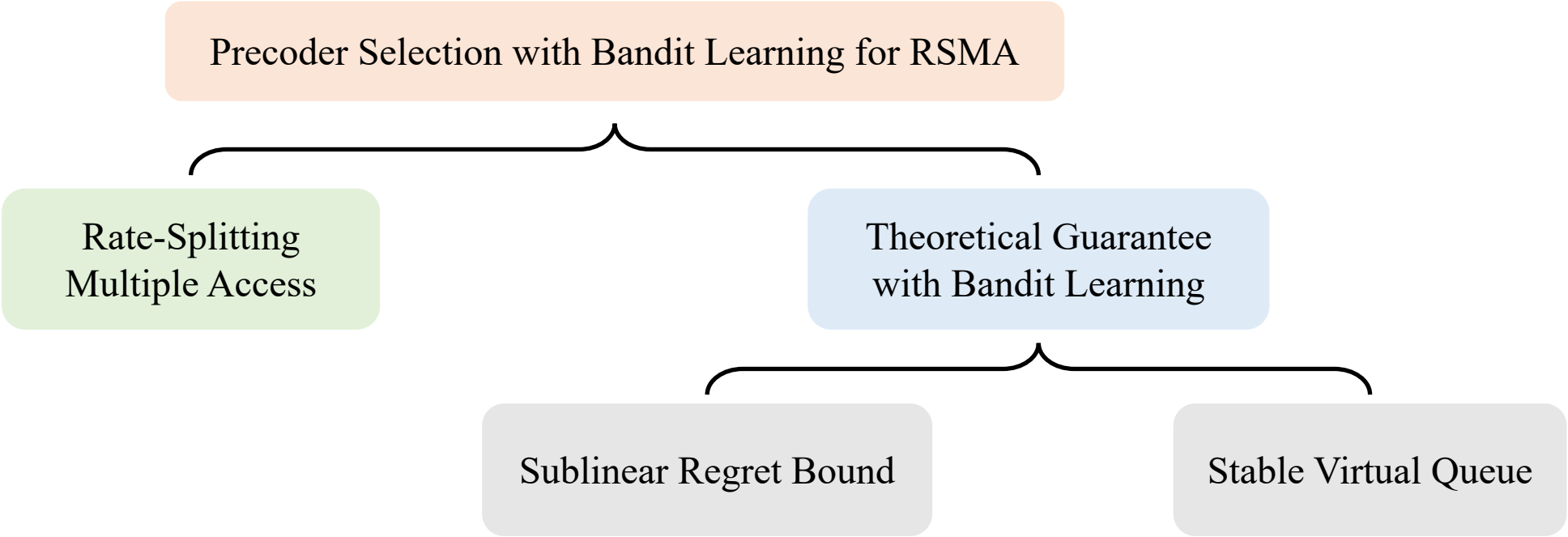
Simulation Result: Power Consumption



Simulation Result: Regret & Power Consumption



Conclusion


Precoder Selection with Bandit Learning for RSMA
Rate-Splitting Multiple Access
Theoretical Guarantee with Bandit Learning
Sublinear Regret Bound
Stable Virtual Queue

Link to our technical report: http://faculty.sist.shanghaitech.edu.cn/faculty/shaozy/ICC23-RSMA.pdf